

NIDHIL NAYUDU

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EDUCATION

University of Michigan

Jan. 2026 – Dec. 2026

Computer Science M.S.E; GPA: 4.00

Ann Arbor, MI

Relevant Coursework: Advanced Scalable Systems, Advanced Compilers, Computer & Network Security, Reinforcement Learning

University of Michigan: 7x Dean's List, 7x University Honors

Aug. 2022 – Dec. 2025

Computer Science B.S.E, Minor in Business; GPA: 3.85

Ann Arbor, MI

Relevant Coursework: Data Structures & Algorithms, Web Systems, Distributed Systems, Operating Systems,

Computer Security, Networking, Machine Learning, Computer Science Theory

EXPERIENCE

Capital One

Starting June 2026

Incoming Software Engineer Intern

New York, NY

CGI

June 2025 - Aug. 2025

Software Development Intern

New York, NY

- Integrated **Model Context Protocol (MCP)** into our **Agentic AI framework** for SDLC automation, decoupling agent logic from client systems to enable server-side updates that **cut client integration time by 42%**.
- Engineered **GitLab CI/CD pipelines** and core Agentic AI agents (BDD test generation/execution, Dev Environment, Pipeline Orchestration), automating ATDD validation and **boosting test coverage by 8%** across over 10 modules.
- Spearheaded **architectural redesign** over three Agile sprints, aligning client and executive goals by leading stakeholder meetings and creating system diagrams that **halved design approval time** and reduced misalignment.

4Human Corporation

May 2024 – Aug. 2024

Software Engineer Intern

New York, NY

- Collaborated with a Ugandan client to streamline **sprint** planning and redesign workflows, **reducing the development cycle time by 15%** for a charitable e-commerce platform.
- Integrated **Firestore** authentication and Firestore database to manage over **50** user, product, and order records in production.
- Refactored over **10** API endpoints into modular Node.js controllers, **improving database query efficiency** at scale.

University of Michigan EECS 280: Programming & Intro Data Structures

Aug. 2024 - Present

Teaching Assistant - Office Hours Lead

Ann Arbor, MI

- Leading lab sections and office hours to assist over **1,000 students**, reinforcing core CS concepts and providing personalized support.
- Coordinating office hours operations for a team of teaching assistants by managing schedules, communicating updates, and organizing additional support during peak office hours with queues of up to 50 students around project deadlines.
- Facilitating understanding of concepts including **recursion** and **polymorphism** by refining lecture material and providing guidance.

PROJECTS

JobAlign: Resume Optimizer - Project Lead | Python, Machine Learning, NLP, RAG, PostgreSQL

- Engineered a custom **SpaCy NLP pipeline** to parse, match, and heuristically optimize resumes for job postings, using a combination of **semantic similarity** and TF-IDF scoring, achieving **75% parsing accuracy**.
- Developed and deployed a Streamlit-based **SaaS** platform for dynamic job matching, using **PostgreSQL as a vector database** to store embeddings and perform top-*k* **semantic search**.
- Implemented a **Retrieval-Augmented Generation (RAG)** system that retrieves the **top 10 most relevant resumes** based on user job input, and integrates them as **context for GPT-4o** to evaluate resume fit and recommend improvements.
- Integrated a chatbot offering users personalized feedback and interactive guidance for efficient recruitment strategies.

ask485?: Search Engine | Python, React.js, Hadoop, Distributed Systems, Flask, Multithreading, APIs

- Created a Google-style scalable search engine with a **React.js** frontend and Python backend to return ranked web search results.
- Built a segmented inverted index using a **MapReduce** pipeline with **Hadoop** streaming for efficient query retrieval.
- Employed **Flask** Python framework to develop **REST APIs** and route requests with backend index and search servers.

Thread Library | C++, Multithreading, Unix, OS, Unit Testing

- Crafted an advanced **C++** thread library with meticulous implementation of **Mutexes**, **Threads**, and **Conditional Variables**, featuring essential functions like **lock**, **unlock**, **signal**, **broadcast**, **wait**, **thread yield**, and **thread join**.
- Implemented **memory management** mechanisms to prevent leaks and enhance library **reliability** in concurrent environments.
- Enforced comprehensive **unit tests** to assure the library's soundness, **identify** and **rectify errors**, and guarantee optimal **performance** in diverse scenarios.

Next Word Prediction Model | Python, Tensorflow, RNNs, NLP, Data Cleaning

- Designed and tuned **RNN architectures** using **LSTM** and **GRU** layers, early stopping, and dropout to reduce overfitting on a **high-performance cluster**.
- Identified limited training data as a bottleneck and pivoted to a 200GB Amazon reviews dataset, boosting accuracy from **25%** to **50%**.
- Deployed a demo with **Streamlit** for showcasing predictions and hyperparameter tuning results interactively.

TECHNICAL SKILLS

Languages : Python, C/C++, Go, JavaScript/TypeScript, HTML/CSS, SQL

Technologies : Next.js/React.js, Git, gdb, Linux, Flask, Multithreading, Networks, AWS, Tensorflow, PyTorch, Docker

Concepts : Object-Oriented Programming, Software Dev, Machine Learning, AI, REST APIs, Agile Dev

Interests : Journaling, Self-learning Telugu, Finance Podcasts, Basketball/Gym, River Tubing